Mechanical Attitude Indication

Introduction:

An artificial horizon, or attitude indicator is a specialized piece of equipment is used by the aerospace industry (and on occasion other industries) to indicate the orientation, with respect to the horizon, (or celestial bodies) of the craft or vessel it is paired with. For a very long time this has been done by reading off the attitude of a specialized gyroscope (typically located behind the instrument) by a complex system of linkages. The gyroscope (powered by compressed air, electric motors or otherwise) would rotate at a programmed pace that would allow the gyroscopic effect to completely counteract any attempted rotation, effectively pinning the gyroscope perpendicular to the gravity plane while the rest of the vessel rotates about it (in a similar vein early guided torpedoes where "aimed" by a gyroscope and pendulum mechanism, allowing them to stay pointed in the right direction). However, due to the linkages between the gyroscope and the indicator, the system can seize and lose any use to the pilot, so in modern times digital screens are used indicate the attitude of a craft, however even this isn't perfect as it uses electricity to run and is therefore susceptible to electronic interface. Thus, I have set out to make a simple device that runs off its own internal grid, is as simple as possible, and gets the job done.

Design:

The flywheel will consist of a $1/8^{th}$ inch thick 3-inch diameter steel disk, radially emerging from the center of the disk, positioned at 90-degree angles from each other and 45-degrees off the disk on both sides, eight pins, each with two small weights act as a form of ballast. When the disk is at rest, the weights shift to the "down" position and move the flywheel assembly's center of mass downwards, but when moving the masses radiate outwards, returning the center of mass to the center of the flywheel. A small dc motor mounted on top of the flywheel will provide the necessary rotation.

Construction:

First the flywheel was constructed with a ¼ inch 3D printed PLA shaft with layer lines running perpendicular to central plate, the central plate fitted was with a ½ inch square hole, interface between plate and center shaft made from a 3/8ths inch tall cuboid with an internal ¼ inch square hole and outer edges that were ½ inch long with 1/8th inch cylindrical cutouts along two opposite edges allowing for two 1/8th inch aluminum rods to be used to both reinforce the core and pin the whole assembly together. Internal components were constructed to transfer shear force to parts with grain structures perpendicular to the shear force. The central system was clasped between two semi circles that were flattened out on the top, and given necessary holes and indentations to allow central shaft and pins to be put thought, additionally four holes angled at 45 degrees from horizontal and placed equidistant from each other were added to the top and bottom retainers to allow for the emplacement of the weights and shafts, terminating in a ring about the top and bottom, the ring used the force of all four rods pushing out on each side to counteract each-other, meaning the only direction they could freely travel in was back into the center, where they were blocked by the core of the flywheel ideally keeping the system in equilibrium.



Clasps had to be added to reinforce connection points. Next, it became necessary to construct a 5-inch sphere able to keep the flywheel rotationally isolated from external rotation. To do this, something similar to a globe was made, eight vertical "ribs" were placed along a ring comprised of eight quarter ring segments, each segment being ¼ of the ring but being bifurcated along the horizontal axis to allow for the lower ring to be rotated 45 degrees, along the center of the total ring, and pinned to the bottom of the ring allowing for all parts to become one cohesive structure, additionally by pinning the "ribs" at either end using a hub of sorts, the resulting force compresses the central ring into something like a single piece. The first attempt quicky broke down due to the "ribs" not actually having anything keeping them in the hubs resulting in them tightening as more "ribs" where added and eventually rebounding and springing out of the structure. In the next iteration, small tabs were added that allowed for complete assembly of the sphere. Lastly, both parts were attached by steel reinforced plastic clips directly connected to the central shaft. On the bottom, the clip terminates directly in the hub with a flange sandwiched between two bearings. On the top, the clip is attached to the shaft of a dc motor which is in turn was firmly mounted into the upper hub.



Testing and comparison:

Testing goes as follows, device was placed in the "vertical" position and voltage was applied to the motor; the result was it attaining a maximum speed just below the speed that the gyroscopic effect occurs. By manually attempting to rotate the device a small amount of procession however minimal. It appears that most of the rotational inertia was being lost to vibration.



Future improvements:

To further minimize vibration, tolerances should be tightened, and the entire central shaft should be fused into a single part. Lastly, a larger motor should be used.

Works cited

"Digitized Documents." U. S. Naval Undersea Museum, 11 Mar. 2026, navalunderseamuseum.org/digitized-documents/.

Sheposh, Richard. *Gyroscope*. Salem Press Encyclopedia of Science, 2026.